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# Dora's Dental Gem - DCMS Build Plan Report
**Purpose:** Break the full Dental Clinic Management System build into
separate, session-sized phases so no single chat/coding session runs out
of usage mid-build, and each phase ends with something testable.

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## Why phase it this way
- Long conversations get expensive fast - every reply re-reads the
growing chat history, so token cost compounds the longer one session
runs.
- Each phase below is scoped to be completable and independently testable
within a single working session, so a session running out mid-task
doesn't leave you with broken, half-wired code.
- Start a **new chat/session** at the start of each phase. Carry forward
only what's needed: the blueprint doc, and a short summary of what was
built in the previous phase.

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## Phase 0 - Setup (small session)
- Initialize project (frontend + backend scaffolding, or Lovable project
if using generated mockups as a base)
- Set up database schema (Patients, Appointments, Treatments, Invoices,
Users/Roles, Notification Log)
- Set up authentication (login, role-based access: Admin / Dentist /
Receptionist)
- **Deliverable:** app boots, login works, roles enforced

## Phase 1 - Patient Management
- Patient registration form
- Patient profile view (bio-data, medical history)
- Patient search/list
- **Deliverable:** can add, view, search, and edit a patient end-to-end

## Phase 2 - Appointment Scheduling
- Calendar/day view
- Book, reschedule, cancel appointment
- Link appointment to a patient record
- **Deliverable:** can book an appointment against a real patient and see
it on the calendar

## Phase 3 - Clinical Records & Treatment
- Dental chart / odontogram (basic version for demo)
- Treatment notes per visit
- Prescriptions
- **Deliverable:** can log a treatment against an appointment/patient

## Phase 4 - Billing & Payments
- Invoice generation from a treatment
- Payment recording (cash/mobile money/etc.)
- Outstanding balance view
- **Deliverable:** can generate and mark an invoice paid

## Phase 5 - Notifications (SMS & Email)
- Notification templates (reminder, confirmation, holiday greeting)
- Trigger appointment confirmation/reminder on booking
- Bulk/broadcast send (holiday message to test list)
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- ****Deliverable:**** booking an appointment triggers a real or logged notification; bulk send works on a test list

Phase 6 – Reports & Admin polish

- Basic dashboard stats (today's appointments, revenue snapshot, patient count)
- Notification delivery log
- Final UI polish/animations pass
- ****Deliverable:**** demo-ready system covering the full client-facing flow

Session-management tips

- One phase = one fresh chat/session where possible. Don't keep stacking phases into one long thread.
- When starting a new phase's session, briefly restate: "Continuing the DCMS build – Phase X is done, here's what exists, now build Phase Y" rather than re-pasting the whole blueprint.
- Prefer ****Claude Code**** (terminal/IDE) over web chat for the actual multi-file coding – it's built for iterative, file-based development and will handle this scale of codebase better than a chat window.
- Check ****Settings → Usage**** in the Claude app before starting a phase if you're unsure how much session budget is left –that's the accurate source, not an estimate.
- If a phase is too large for one session (Phase 5 or 6 might be), split further: e.g. Phase 5a (templates + triggers) and Phase 5b (bulk send + delivery log).

Suggested order rationale

Phases follow the same dependency chain as the data itself: you need a patient before an appointment, an appointment before a treatment, a treatment before an invoice, and a booked appointment before a notification has anything to reference. Building in this order means every phase produces a working, demo-able slice rather than disconnected pieces.